

Financial Credit Transaction Analysis

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Problem Statement :

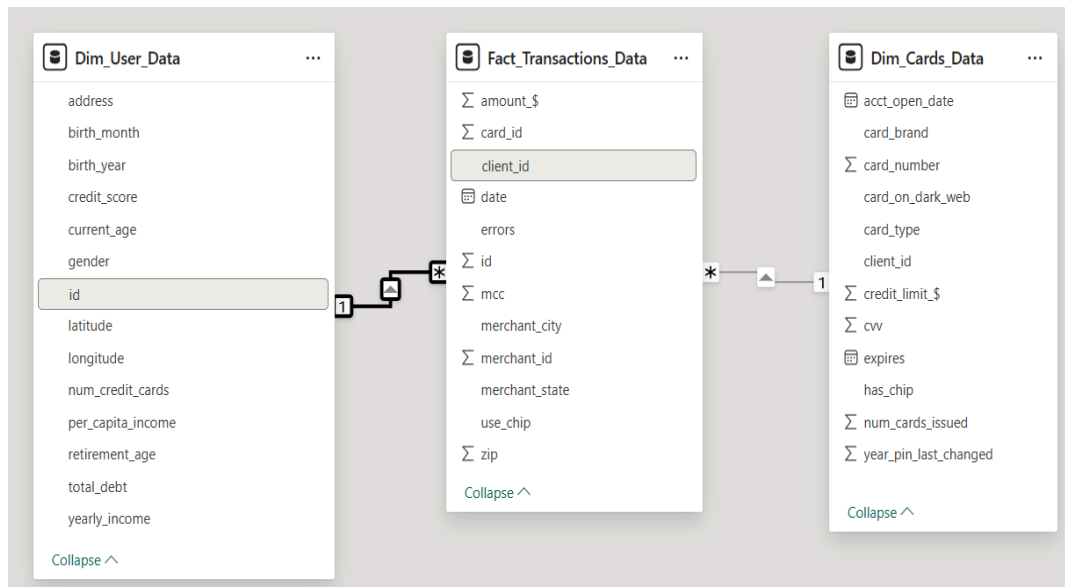
This star schema functions as a financial decision engine, converting raw transactions into strategic foresight. By correlating spending patterns with user health (DTI and CUR) and security flags, you can proactively mitigate risk, automate fraud detection, and personalize growth strategies. Ultimately, the model shifts your focus from descriptive reporting to predictive action.

Data Collection:

Data Source: Kaggle

This comprehensive financial dataset combines transaction records, customer information, and card data from a banking institution, spanning the 2010s decade. The dataset is designed for multiple analytical purposes, including synthetic fraud detection, customer behavior analysis, and expense forecasting.

Data Analysis & Modeling:



User Data :

	id	current_age	retirement_age	birth_year	birth_month	gender	address	latitude	longitude	per_capita_income	yearly_income	total_debt	credit_score	num_credit_cards
▶	825	53	66	1966	11	Female	462 Rose Lane	34.15	-117.76	\$29278	\$59696	\$127613	787	5
	1746	53	68	1966	12	Female	3606 Federal Boulevard	40.76	-73.74	\$37891	\$77254	\$191349	701	5
	1718	81	67	1938	11	Female	766 Third Drive	34.02	-117.89	\$22681	\$33483	\$196	698	5
	708	63	63	1957	1	Female	3 Madison Street	40.71	-73.99	\$163145	\$249925	\$202328	722	4
	1164	43	70	1976	9	Male	9620 Valley Stream Drive	37.76	-122.44	\$53797	\$109687	\$183855	675	1
	68	42	70	1977	10	Male	58 Birch Lane	41.55	-90.6	\$20599	\$41997	\$0	704	3
	1075	36	67	1983	12	Female	5695 Fifth Street	38.22	-85.74	\$25258	\$51500	\$102286	672	3
	1711	26	67	1993	12	Male	1941 Ninth Street	45.51	-122.64	\$26790	\$54623	\$114711	728	1
	1116	81	66	1938	7	Female	11 Spruce Avenue	40.32	-75.32	\$26273	\$42509	\$2885	755	5
	1752	34	60	1986	1	Female	887 Grant Street	29.97	-92.12	\$18730	\$38190	\$2895	810	1
	192	27	66	1992	6	Male	888 Fifth Lane	38.65	-121.25	\$27548	\$56164	\$15224	761	2
	640	29	63	1990	9	Female	8677 Littlewood Lane	40.42	-104.74	\$22427	\$45727	\$94016	629	1

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Card Data :

	id	client_id	card_brand	card_type	card_number	expires	cvv	has_chip	num_cards_issued	credit_limit	acct_open_date	year_pin_last_changed	card_on_dark_web
▶	4524	825	Visa	Debit	4344676511950444	12/2022	623	YES	2	\$24295	09/2002	2008	No
	2731	825	Visa	Debit	4956965974959986	12/2020	393	YES	2	\$21968	04/2014	2014	No
	3701	825	Visa	Debit	4582313478255491	02/2024	719	YES	2	\$46414	07/2003	2004	No
	42	825	Visa	Credit	4879494103069057	08/2024	693	NO	1	\$12400	01/2003	2012	No
	4659	825	Mastercard	Debit (Prepaid)	5722874738736011	03/2009	75	YES	1	\$28	09/2008	2009	No
	4537	1746	Visa	Credit	4404898874682993	09/2003	736	YES	1	\$27500	09/2003	2012	No
	1278	1746	Visa	Debit	4001482973848631	07/2022	972	YES	2	\$28508	02/2011	2011	No
	3687	1746	Mastercard	Debit	5627220683410948	06/2022	48	YES	2	\$9022	07/2003	2015	No
	3465	1746	Mastercard	Debit (Prepaid)	5711382187309326	11/2020	722	YES	2	\$54	06/2010	2015	No
	3754	1746	Mastercard	Debit (Prepaid)	5766121508358701	02/2023	908	YES	1	\$99	07/2006	2012	No
	5144	1718	Mastercard	Debit	5495199163052054	03/2022	677	YES	2	\$31599	10/2009	2009	No
	2029	1718	Mastercard	Debit	5804499644308599	07/2023	258	NO	2	\$27480	03/2002	2008	No
	2379	1718	Mastercard	Debit	5766352389579834	02/2020	992	YES	1	\$26743	03/2019	2019	No
	2732	1718	Visa	Debit	4242015583697294	06/2020	928	YES	1	\$31463	04/2014	2014	No
	4706	1718	Mastercard	Debit	5191030913182493	06/2024	360	YES	1	\$16055	09/2009	2009	No

Transactions Data :

	id	date	client_id	card_id	amount	use_chip	merchant_id	merchant_city	merchant_state	zip	mcc	errors
▶	7475327	2010-01-01 00:01:00	1556	2972	\$-77.00	Swipe Transaction	59935	Beulah	ND	58523	5499	
	7475328	2010-01-01 00:02:00	561	4575	\$14.57	Swipe Transaction	67570	Bettendorf	IA	52722	5311	
	7475329	2010-01-01 00:02:00	1129	102	\$80.00	Swipe Transaction	27092	Vista	CA	92084	4829	
	7475331	2010-01-01 00:05:00	430	2860	\$200.00	Swipe Transaction	27092	Crown Point	IN	46307	4829	
	7475332	2010-01-01 00:06:00	848	3915	\$46.41	Swipe Transaction	13051	Harwood	MD	20776	5813	
	7475333	2010-01-01 00:07:00	1807	165	\$4.81	Swipe Transaction	20519	Bronx	NY	10464	5942	
	7475334	2010-01-01 00:09:00	1556	2972	\$77.00	Swipe Transaction	59935	Beulah, Bronx	ND	58523	5499	
	7475337	2010-01-01 00:21:00	351	1112	\$10.74	Swipe Transaction	3864	Flushing	NY	11355	5813	
	7475338	2010-01-01 00:23:00	554	3912	\$3.51	Swipe Transaction	67570	Pearland	TX	77581	5311	
	7475339	2010-01-01 00:23:00	605	5061	\$2.58	Swipe Transaction	75781	Brooklyn	NY	11210	5411	
	7475340	2010-01-01 00:26:00	1556	2972	\$39.63	Swipe Transaction	59935	Beulah	ND	58523	5499	
	7475341	2010-01-01 00:27:00	1797	1127	\$43.33	Swipe Transaction	33326	Kahului	HI	96732	4121	
	7475342	2010-01-01 00:30:00	114	3398	\$49.42	Swipe Transaction	61195	North Hollyw...	CA	91606	5541	

Insights :

Que 1: Get the number of data points for each table

```

15 • select
16     (select count(distinct id) from cards_data) as Card_Datapoints,
17     (select count(distinct id) from transactions_data) as Transaction_DataPoints,
18     (select count(distinct id) from users_data) as User_DataPoints;
19 -----

```

result Grid	Filter Rows:	Export:	Wrap Cell Content:
Card_Datapoints	Transaction_DataPoints	User_DataPoints	
6146	40911	2000	

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Now I want to analyze Table:

User Data :

Que 1.1: Categorize the users based on the Credit Score, and get the number of users

```
select
case
    when credit_score > 800 then "Exceptional"
    when credit_score > 740 then "Very Good"
    when credit_score > 670 then "Good"
    when credit_score > 580 then "Fair"
    when credit_score > 300 then "Poor"
end as User_type, count(*) as "Number of User" from users_data
group by User_type;
```

	User_type	Number of User
▶	Very Good	468
	Good	937
	Exceptional	163
	Fair	350
	Poor	82

Que 1.2 : Find individuals who have stopped using their credit cards.

```
SELECT * FROM users_data
WHERE id NOT IN (SELECT DISTINCT client_id FROM transactions_data);

SELECT
    users_data.id, users_data.gender
FROM
    users_data
    LEFT JOIN
    transactions_data ON users_data.id = transactions_data.client_id
WHERE
    transactions_data.client_id IS NULL;
```

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Que 1.3: Identify the user who is both a high-frequency credit card user and a high-risk client.

Note : Aim to keep Credit Utilization below 30%

Credit Utilization = (total_Spend / total_limit) * 100

Zone	Utilization Range	Implication
Safe Zone	0% - 10%	Optimal usage.
Warning Zone	10% - 30%	Approaching higher risk.
Stress Zone	>30%	High-risk usage; generally undesirable.

The solution can be divided into four distinct steps:

1. Calculate the total aggregate spending.
2. Determine the overall credit limit.
3. Compute the credit utilization ratio (index).
4. Identify the user's appropriate tier based on the computed utilization ratio.

Part 01 : Total Spending

```
# Part 01: Calculate the total aggregate spending.
SELECT
    u.id,
    u.retirement_age,
    u.credit_score,
    SUM(CAST(REPLACE(t.amount, '$', '') AS DECIMAL (15 , 2 ))) AS Total_Transaction_Amount
FROM
    users_data u
    LEFT JOIN
    transactions_data t ON u.id = t.client_id
GROUP BY u.id , u.retirement_age , u.credit_score;
```

id	retirement_age	credit_score	Total_Transaction_Amount
825	66	787	2915.65
1746	68	701	1333.70
1718	67	698	746.41
708	63	722	3527.48
1164	70	675	3230.53
68	70	704	1577.06
1075	67	672	2121.97
1116	66	755	3073.95
1752	60	810	1356.89

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Part 02 : Total Credit Limit

Part 02 : Determine the overall credit limit.

```
SELECT
    c.client_id,
    SUM(c.num_cards_issued) AS 'Total Cards',
    SUM(CAST(REPLACE(c.credit_limit, '$', '') AS DECIMAL (15 , 2 ))) AS 'Credit_Limit'
FROM
    cards_data c
GROUP BY c.client_id;
```

client_id	Total Cards	Credit_Limit
825	8	105105.00
1746	8	65183.00
1718	7	133340.00
708	5	356324.00
1164	1	34900.00
68	6	47287.00
1075	5	67917.00
1711	2	17480.00
1116	7	103932.00

Now it's time to find out Card Utilization Score :

```
select total_transaction_table.id,
       round((total_transaction_table.total_transaction_amount/Credit_limit.Credit_Limit)*100,2) 'Credit Utilization Score'
from
    ⊕ (SELECT
        total_transaction_table
    ⊕ left join (
        Credit_limit
    on Credit_limit.client_id = total_transaction_table.id;
```

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	id	Credit Utilization Score
▶	825	2.77
	1746	2.05
	1718	0.56
	708	0.99
	1164	9.26
	68	3.34
	1075	3.12
	1116	2.96

Que 1.4: Debt-to-Income (DTI) Ratio Guidelines and Query Objective

The following guidelines apply to the Debt-to-Income (DTI) ratio:

- **Favorable:** A DTI ratio below 40% (0.4).
- **High-Risk:** A DTI ratio exceeding 40% (0.4).

Assumption: The analysis of this dataset is based on a one-year time horizon.

Formula :

$$\text{DTI Ratio} = (\text{Total Monthly Debt Payments} \div \text{Gross Monthly Income}) \times 100$$

In our case :

$$\text{DTI Ratio} = (\text{Total Yearly Debt Payments} \div \text{Gross Yearly Income}) \times 100$$

Query Objective: We seek to identify users who meet the criteria of having a high Credit Score (greater than 700) *and* a favorable DTI Ratio (less than 0.4). A query should be written to select these users.

```
SELECT
    id, current_age, gender, retirement_age, per_capita_income
FROM
    users_data
WHERE
    credit_score > 700
    AND (0.4 > (RIGHT(total_debt,
        LENGTH(total_debt) - 1) / RIGHT(yearly_income,
        LENGTH(yearly_income) - 1)));
```

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	id	current_age	gender	retirement_age	per_capita_income
	1227	65	Female	67	\$33031
	454	88	Female	65	\$22214
	108	50	Female	68	\$18709
	1888	75	Female	59	\$28429
	1276	82	Female	66	\$25365
	851	88	Female	65	\$19620
	1951	48	Female	66	\$31075
	339	19	Female	75	\$15597
	736	27	Female	66	\$30072